

(Two Years Programme) **(Batch 2024-26)**
(2224)

Paper : CSL-4036 : Machine Learning**Time allowed: 3 hrs.****Max. Marks: 70**

Note: All questions carry equal marks. Candidates are required to attempt five questions, selecting at least one question from each Section. The fifth question may be attempted from any Section.

SECTION A

- Q1. What are the main challenges faced in machine learning? Explain how a confusion matrix is used in evaluating the performance of an ML model. (14)
- Q2. Discuss the concepts of probability and classification in ML. Explain Bayes optimal decisions and the Naive Bayes classifier. (14)

SECTION B

- Q3. Describe linear regression, including the concepts of sum of squared errors (SSE), gradient descent, and normal equations. How can overfitting and underfitting be addressed? (14)
- Q4. Explain the concept of logistic regression and contrast it with linear and non-linear SVM classifications in terms of decision boundaries and suitability for different types of data. (14)

SECTION C

- Q5. Discuss the curse of dimensionality and its impact on machine learning. Explain dimensionality reduction techniques, focusing on Principal Components Analysis (PCA). (14)
- Q6. Define unsupervised learning and its significance. Explain various clustering algorithms and their applications in image segmentation and data preprocessing. (14)

SECTION D

- Q7. Explain the fundamentals of reinforcement learning. Discuss the model-based reinforcement learning paradigm, and elaborate on the role of Q-learning as a key algorithm within this framework. (14)
- Q8. Discuss the diverse applications of reinforcement learning across different domains. Illustrate your answer with examples from areas such as robotics, gaming, and any other relevant fields, highlighting how reinforcement learning enhances performance in these scenarios. (14)
